Q.23 Differentiate between types of plastics? (CO2)

Q.24 Write a short note on semiconductor materials? (CO2)

Q.25 Briefly explain cooling curve of a pure metal? (CO2)

Q.26 Explain lever rule with diagram? (CO2)

Q.27 Draw basic process of iron and steel making. (CO2)

Q.28 What are the effects of carbon and tungsten in steel (CO2)

Q.29 What is the chemical composition and properties of high speed steel? (CO2)

Q.30 What are the objectives of quenching ? (CO3)

Q.31 Explain any one type of heat treatment furnace? (CO4)

Q.32 What are the properties of thermoset plastics?

Q.33 Write a note on food grade plastics? (CO2)

Q.34 Write a note on smart materials in 100 words? (CO4)

Q.35 Describe the properties of copper.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Draw a neat sketch of Iron carbon diagram and Explain it.

Q.37 Explain TTT diagram? (CO5)

Q.38 Briefly describe various mechanical properties? (CO2)

(**Note:** Course outcome/CO is for office use only)

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4th Sem / Auto, Mech, Prod, T&D, GE, CAD/CAM, CNC, Metallurgy, Found. & Forg, Adv. Manuf. Tech., Mech Engg (Fabrication Tech), Mech Engg (CAD/ CAM Dsgn & Robotics)

Subject:- Materials and Metallurgy / Met. Sci.

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 What is the coordination number of a face centered cubic (FCC) unit cell?

a) 4 b) 6

c) 8 d) 12

Q.2 Effective number of atoms in a face centered cubic (FCC) unit cell is equal to________

a) 4 b) 1

c) 8 d) 2

Q.3 The atomic packing fraction in a body centered cubic unit cell is________

a) 0.74 b) 0.52

c) 0.68 d) 0.66

Q.4 Cast iron is a product of_________

a) Cupola b) Bessemer converter

c) Open hearth furnace d) Blast furnace

Q.5 Red hardness of alloy steel can be improved by adding_______

a) Manganese b) Vanadium

c) Tungsten d) Titanium

Q.6 In Annealing, cooling is done in which of the following medium?

a) Air b) Water

c) Oil d) Furnace

Q.7 Mild steel can be converted into high carbons steel by which of the following heat treatment process?

a) Annealing b) Normalizing

c) Case hardening d) Nitriding

Q.8 Which of the following hardness test uses steel ball as indenter?

a) Brinell hardness test

b) Rockwell C hardness test

c) Vickers hardness test

d) Rockwell B hardness test

Q.9 Substitution of a foreign atom in the site of parent atom in the crystal is a ?

a) Vacancy defect

b) Substitution impurity

c) Volume imperfection

d) Vacancy defect

Q.10 As the grain size of a metal increases, its strength _________

a) Decreases

b) Increases

c) Remains constant

d) No effect of grain size on strength

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 All engineering materials are plastic. (True/False)

Q.12 Melting point of iron (in °C) is _________

Q.13 Major constituents of the gun metal is_________.

Q.14 The composite constituents of both matrix and reinforcement are softer. (True/False)

Q.15 Name any two spring materials?

Q.16 Write composition of bronze?

Q.17 Name any one heat insulating material?

Q.18 Cast iron is a ductile material.(True/False)

Q.19 Cast iron contain maximum of______% carbon.

Q.20 Strength of ductile materials is find out by_________ test.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Differentiate between metals and non-metals? (CO1)

Q.22 Derive atomic packaging factor of simple cubic crystal? (CO4)